

# SOLIDWORKS API (.NET VBA C#) — Reference (267/337): ISketchBlockDefinition

SOLIDWORKS API Documentation (converted from CHM)

# ISketchBlockDefinition (Type)

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## ISketchBlockDefinition Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchBlockDefinition Interface

Allows access to information about a block definition.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface ISketchBlockDefinition
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition
```

C#

```
public interface ISketchBlockDefinition
```

C++/CLI

```
public interface class ISketchBlockDefinition
```

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition](#).

### Example

[Get and Set Blocks Data in Any Document \(VBA\)](#)  
[Get Block Instance in Part or Assembly \(VBA\)](#)  
[Get Blocks in Drawing \(VBA\)](#)  
[Create Block Definition and Insert Block Instance \(C#\)](#)  
[Create Block Definition and Insert Block Instance \(VBA\)](#)  
[Create Block Definition and Insert Block Instance \(VB.NET\)](#)

### Remarks

Using this interface, you can:

- Get and set:
  - Name of the block
  - Whether the block links to a file, and get or set the name of that file
- Get:
  - All [instances of the block](#)
  - Sketch of this block definition
  - All notes and dimensions and the number of all notes and dimensions

See [Block Definitions and Block Instances](#) for details.

### Accessors

[IFeature::GetSpecificFeature2](#)  
[ISketchBlockInstance::Definition](#)  
[ISketchManager::GetSketchBlockDefinitions](#) and [ISketchManager::IGetSketchBlockDefinitions](#)  
[ISketchManager::MakeSketchBlockFromFile](#)  
[ISketchManager::MakeSketchBlockFromSelected](#)  
[ISketchManager::MakeSketchBlockFromSketch](#)

## Access Diagram

[SketchBlockDefinition](#)

## See Also

[ISketchBlockDefinition Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISketchManager Interface](#)  
[ISketchBlockInstance Interface](#)

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## ISketchBlockDefinition Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchBlockDefinition Interface

The following tables list the members exposed by [ISketchBlockDefinition](#).

## Public Properties

| Name                                                                                                             | Description                                                                |
|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|  <a href="#">FileName</a>       | Gets or sets the name of the file to which the block definition is linked. |
|  <a href="#">InsertionPoint</a> | Gets or sets the insertion point for the block definition.                 |
|  <a href="#">LinkToFile</a>     | Gets or sets whether the block definition file is linked to a file.        |

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## Public Methods

| Name                                                                                                                          | Description                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">GetArcCount</a>                | Gets the number of arcs in this block definition.                                                                                                                                                                 |
|  <a href="#">GetArcs</a>                   | Gets information about all of the arcs in the block definition.                                                                                                                                                   |
|  <a href="#">GetDisplayDimensionCount</a>  | Gets the number of display dimensions in this block definition.                                                                                                                                                   |
|  <a href="#">GetDisplayDimensions</a>      | Gets the display dimensions in this block definition.                                                                                                                                                             |
|  <a href="#">GetEllipseCount</a>           | Gets the number of ellipses in this block definition.                                                                                                                                                             |
|  <a href="#">GetEllipses</a>               | Gets the information about all of the ellipses in this block definition.                                                                                                                                          |
|  <a href="#">GetFeature</a>                | Gets the feature for this block definition.                                                                                                                                                                       |
|  <a href="#">GetInstanceCount</a>          | Gets the number of block instances of this block definition.                                                                                                                                                      |
|  <a href="#">GetInstances</a>              | Gets all of the block instances of this block definition.                                                                                                                                                         |
|  <a href="#">GetLineCount</a>              | Gets the number of lines in this block definition.                                                                                                                                                                |
|  <a href="#">GetLines</a>                  | Gets information about all of the lines in this block definition.                                                                                                                                                 |
|  <a href="#">GetNoteCount</a>              | Gets the number of notes in this block definition.                                                                                                                                                                |
|  <a href="#">GetNotes</a>                  | Gets the notes in this block definition.                                                                                                                                                                          |
|  <a href="#">GetParabolaCount</a>          | Gets the number of parabolas in this block definition.                                                                                                                                                            |
|  <a href="#">GetParabolas</a>              | Gets information about all of the parabolas in this block definition.                                                                                                                                             |
|  <a href="#">GetSketch</a>                 | Gets the underlying sketch of this block definition.                                                                                                                                                              |
|  <a href="#">GetSplineCount</a>            | Gets the number of splines in this block definition.                                                                                                                                                              |
|  <a href="#">GetSplineInterpolateCount</a> | Gets the number of splines in the sketch block definition and the size of array required to hold the data for the interpolation of these splines.                                                                 |
|  <a href="#">GetSplineParams</a>           | Gets all the parameters of the splines in the sketch block definition.                                                                                                                                            |
|  <a href="#">GetSplineParamsCount</a>      | Gets the number of splines in the sketch block definition and the size of array required to hold the data for the parameters of these splines.                                                                    |
|  <a href="#">GetSplines</a>                | Obsolete. Superseded by <a href="#">ISketchBlockDefinition.GetSplines2</a> and <a href="#">ISketchBlockDefinition.IGetSplines2</a> .                                                                              |
|  <a href="#">GetSplines2</a>               | Gets the spline points by tessellation instead of by interpolation as is done by <a href="#">ISketchBlockDefinition.GetSplinesInterpolate</a> and <a href="#">ISketchBlockDefinition.IGetSplinesInterpolate</a> . |
|  <a href="#">GetSplinesInterpolate</a>     | Gets all of the parameters of the splines by interpolation instead of by tessellation as is done by <a href="#">ISketch.GetSplines2</a> and <a href="#">ISketch.IGetSplines2</a> .                                |

|                                                                                   |                                        |                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                                        |                                                                                                                                                                                                                   |
|  | <a href="#">GetUserPoints</a>          | Gets information about all of the user points in this block definition.                                                                                                                                           |
|  | <a href="#">GetUserPointsCount</a>     | Gets the number of user points in this block definition.                                                                                                                                                          |
|  | <a href="#">IGetArcs</a>               | Gets information about all of the arcs in the block definition.                                                                                                                                                   |
|  | <a href="#">IGetDisplayDimensions</a>  | Gets the display dimensions in this block definition.                                                                                                                                                             |
|  | <a href="#">IGetEllipses</a>           | Gets the information about all of the ellipses in this block definition.                                                                                                                                          |
|  | <a href="#">IGetInstances</a>          | Gets all of the block instances of this block definition.                                                                                                                                                         |
|  | <a href="#">IGetLines</a>              | Gets information about all of the lines in this block definition.                                                                                                                                                 |
|  | <a href="#">IGetNotes</a>              | Gets the notes in this block definition.                                                                                                                                                                          |
|  | <a href="#">IGetParabolas</a>          | Gets information about all of the parabolas in this block definition.                                                                                                                                             |
|  | <a href="#">IGetSplineParams</a>       | Gets all the parameters of the splines in the sketch block definition.                                                                                                                                            |
|  | <a href="#">IGetSplines</a>            | <b>Obsolete. Superseded by <a href="#">ISketchBlockDefinition.GetSplines2</a> and <a href="#">ISketchBlockDefinition.IGetSplines2</a></b>                                                                         |
|  | <a href="#">IGetSplines2</a>           | Gets the spline points by tessellation instead of by interpolation as is done by <a href="#">ISketchBlockDefinition.GetSplinesInterpolate</a> and <a href="#">ISketchBlockDefinition.IGetSplinesInterpolate</a> . |
|  | <a href="#">IGetSplinesInterpolate</a> | Gets all of the paramereers of the splines by interpolation instead of by tessellation as is done by <a href="#">ISketch.GetSplines2</a> and <a href="#">ISketch.IGetSplines2</a> .                               |
|  | <a href="#">IGetUserPoints</a>         | Gets information about all of the user points in this block definition.                                                                                                                                           |
|  | <a href="#">Save</a>                   | Saves the block definition.                                                                                                                                                                                       |

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## See Also

[ISketchBlockDefinition Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISketchManager Interface](#)  
[ISketchBlockInstance Interface](#)

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## ISketchBlockDefinition Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchBlockDefinition Interface

For a list of all members of this type, see [ISketchBlockDefinition members](#).

## Public Methods

| Name                                                                                                                         | Description                                                              |
|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
|  <a href="#">GetArcCount</a>              | Gets the number of arcs in this block definition.                        |
|  <a href="#">GetArcs</a>                  | Gets information about all of the arcs in the block definition.          |
|  <a href="#">GetDisplayDimensionCount</a> | Gets the number of display dimensions in this block definition.          |
|  <a href="#">GetDisplayDimensions</a>     | Gets the display dimensions in this block definition.                    |
|  <a href="#">GetEllipseCount</a>          | Gets the number of ellipses in this block definition.                    |
|  <a href="#">GetEllipses</a>              | Gets the information about all of the ellipses in this block definition. |
|  <a href="#">GetFeature</a>               | Gets the feature for this block definition.                              |
|  <a href="#">GetInstanceCount</a>         | Gets the number of block instances of this block definition.             |
|  <a href="#">GetInstances</a>             | Gets all of the block instances of this block definition.                |
|  <a href="#">GetLineCount</a>             | Gets the number of lines in this block definition.                       |
|  <a href="#">GetLines</a>                 | Gets information about all of the lines in this block definition.        |
|  <a href="#">GetNoteCount</a>             | Gets the number of notes in this block definition.                       |
|  <a href="#">GetNotes</a>                 | Gets the notes in this block definition.                                 |
|  <a href="#">GetParabolaCount</a>         | Gets the number of parabolas in this block definition.                   |
|  <a href="#">GetParabolas</a>             | Gets information about all of the parabolas in this block definition.    |

|                                                                                     |                                           |                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <a href="#">GetSketch</a>                 | Gets the underlying sketch of this block definition.                                                                                                                                                                |
|    | <a href="#">GetSplineCount</a>            | Gets the number of splines in this block definition.                                                                                                                                                                |
|    | <a href="#">GetSplineInterpolateCount</a> | Gets the number of splines in the sketch block definition and the size of array required to hold the data for the interpolation of these splines.                                                                   |
|    | <a href="#">GetSplineParams</a>           | Gets all the parameters of the splines in the sketch block definition.                                                                                                                                              |
|    | <a href="#">GetSplineParamsCount</a>      | Gets the number of splines in the sketch block definition and the size of array required to hold the data for the parameters of these splines.                                                                      |
|    | <a href="#">GetSplines</a>                | <b>Obsolete. Superseded by <a href="#">ISketchBlockDefinition::GetSplines2</a> and <a href="#">ISketchBlockDefinition::IGetSplines2</a></b>                                                                         |
|    | <a href="#">GetSplines2</a>               | Gets the spline points by tessellation instead of by interpolation as is done by <a href="#">ISketchBlockDefinition::GetSplinesInterpolate</a> and <a href="#">ISketchBlockDefinition::IGetSplinesInterpolate</a> . |
|    | <a href="#">GetSplinesInterpolate</a>     | Gets all of the parameters of the splines by interpolation instead of by tessellation as is done by <a href="#">ISketch::GetSplines2</a> and <a href="#">ISketch::IGetSplines2</a> .                                |
|    | <a href="#">GetUserPoints</a>             | Gets information about all of the user points in this block definition.                                                                                                                                             |
|    | <a href="#">GetUserPointsCount</a>        | Gets the number of user points in this block definition.                                                                                                                                                            |
|    | <a href="#">IGetArcs</a>                  | Gets information about all of the arcs in the block definition.                                                                                                                                                     |
|    | <a href="#">IGetDisplayDimensions</a>     | Gets the display dimensions in this block definition.                                                                                                                                                               |
|    | <a href="#">IGetEllipses</a>              | Gets the information about all of the ellipses in this block definition.                                                                                                                                            |
|    | <a href="#">IGetInstances</a>             | Gets all of the block instances of this block definition.                                                                                                                                                           |
|    | <a href="#">IGetLines</a>                 | Gets information about all of the lines in this block definition.                                                                                                                                                   |
|    | <a href="#">IGetNotes</a>                 | Gets the notes in this block definition.                                                                                                                                                                            |
|    | <a href="#">IGetParabolas</a>             | Gets information about all of the parabolas in this block definition.                                                                                                                                               |
|    | <a href="#">IGetSplineParams</a>          | Gets all the parameters of the splines in the sketch block definition.                                                                                                                                              |
|    | <a href="#">IGetSplines</a>               | <b>Obsolete. Superseded by <a href="#">ISketchBlockDefinition::GetSplines2</a> and <a href="#">ISketchBlockDefinition::IGetSplines2</a></b>                                                                         |
|    | <a href="#">IGetSplines2</a>              | Gets the spline points by tessellation instead of by interpolation as is done by <a href="#">ISketchBlockDefinition::GetSplinesInterpolate</a> and <a href="#">ISketchBlockDefinition::IGetSplinesInterpolate</a> . |
|    | <a href="#">IGetSplinesInterpolate</a>    | Gets all of the parameters of the splines by interpolation instead of by tessellation as is done by <a href="#">ISketch::GetSplines2</a> and <a href="#">ISketch::IGetSplines2</a> .                                |
|   | <a href="#">IGetUserPoints</a>            | Gets information about all of the user points in this block definition.                                                                                                                                             |
|  | <a href="#">Save</a>                      | Saves the block definition.                                                                                                                                                                                         |

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## See Also

[ISketchBlockDefinition Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISketchManager Interface](#)  
[ISketchBlockInstance Interface](#)

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## ISketchBlockDefinition Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchBlockDefinition Interface

For a list of all members of this type, see [ISketchBlockDefinition members](#).

## Public Properties

| Name                                                                                                               | Description                                                                |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|  <a href="#">FileName</a>       | Gets or sets the name of the file to which the block definition is linked. |
|  <a href="#">InsertionPoint</a> | Gets or sets the insertion point for the block definition.                 |
|  <a href="#">LinkToFile</a>     | Gets or sets whether the block definition file is linked to a file.        |

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## See Also

[ISketchBlockDefinition Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISketchManager Interface](#)  
[ISketchBlockInstance Interface](#)

## FileName Property (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : FileName Property (ISketchBlockDefinition)

Gets or sets the name of the file to which the block definition is linked.

### .NET Syntax

#### Visual Basic (Declaration)

Property FileName As System.String

#### Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition
Dim value As System.String

instance.FileName = value

value = instance.FileName
```

#### C#

```
System.string FileName {get; set;}
```

#### C++/CLI

```
property System.String^ FileName {
    System.String^ get();
    void set ( & System.String^ value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Path and filename of the block definition file

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::FileName](#).

### Remarks

The file can be either a native SOLIDWORKS block file (.sldblk) or a non-native SOLIDWORKS file (.dxf or .dwg).

This method gets the name of the file with which this block definition is associated, regardless of whether or not the link is enabled. SOLIDWORKS continues to store the name of the file if the link is disabled. Use [ISketchBlockDefintion::LinkToFile](#) to determine if the block definition is linked to a file.

### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::Save Method](#)

### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

## GetArcCount Method (ISketchBlockDefinition)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetArcCount Method (ISketchBlockDefinition)

Gets the number of arcs in this block definition.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetArcCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Integer  
  
value = instance.GetArcCount()
```

C#

```
System.int GetArcCount()
```

C++/CLI

```
System.int GetArcCount();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Number of arcs

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetArcCount](#).

## Example

[Get Block Instance in Part or Assembly \(VBA\)](#)

## Remarks

Call this method before calling [ISketchBlockDefinition::IGetArcs](#) to get the size of the array for that method.

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::IGetArcs Method](#)

## Availability

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## GetArcs Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetArcs Method (ISketchBlockDefinition)

Gets information about all of the arcs in the block definition.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetArcs() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Object
```

```
value = instance.GetArcs()
```

C#

```
System.Object GetArcs()
```

C++/CLI

```
System.Object^ GetArcs();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array containing an array of doubles (see **Remarks**)

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetArcs](#).

### Remarks

Return value is an array of doubles with the format:

[ *Color*, *Type*, *Line Font*, *Line Width*, *Layer ID*, *Layer Override*, [*Start*], [*End*], [*Center*] *Rotation Dir* ]

where

- *Color* - COLORREF returned as an integer. Return value could be 0 or -1 for default color.
- *Type* - Line type. Valid returns as defined in [swLineTypes\\_e](#). A LineType is a combination of a line style and line weight.
- *Line Font* - Line style. Valid line styles as defined in [swLineStyles\\_e](#) enumeration.
- *Line Width* - Integer value defining the line width. Valid width values as defined in [swLineWeights\\_e](#).
- *Layer ID* - Integer value indicating which layer holds this entity. The [ILayer](#) object can be obtained by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::IGetLayerID](#).
- *Layer Override* - Integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, the color and style have been specifically set for this item and may not match the default values associated with this item's layer.
- *Start[3]* - An array of 3 doubles (X,Y,Z) describing the start point.
- *End[3]* - An array of 3 doubles (X,Y,Z) describing the end point. If the arc is closed, then End[3] is the same point as Start.
- *Center[3]* - An array of 3 doubles (X,Y,Z) describing the center point.
- *Rotation Dir* - Rotational direction (CW = -1, CCW = 1)

This set of data repeats for each arc in the sketch. The size of the array is (NumArcs \* 16).

### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::IGetArcs Method](#)  
[ISketchBlockDefinition::GetArcCount Method](#)

### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## GetDisplayDimensionCount Method (ISketchBlockDefinition)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetDisplayDimensionCount Method (ISketchBlockDefinition)

Gets the number of display dimensions in this block definition.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetDisplayDimensionCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition
Dim value As System.Integer

value = instance.GetDisplayDimensionCount()
```

C#

```
System.int GetDisplayDimensionCount()
```

C++/CLI

```
System.int GetDisplayDimensionCount();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Number of display dimensions

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetDisplayDimensionCount](#).

### Example

[Get Block Instance in Part or Assembly \(VBA\)](#)

### Remarks

Call this method before calling [ISketchBlockDefinition::IGetDisplayDimensions](#) to get the size of the array for that method.

### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetDisplayDimensions Method](#)

### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## GetDisplayDimensions Method (ISketchBlockDefinition)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetDisplayDimensions Method (ISketchBlockDefinition)

Gets the display dimensions in this block definition.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetDisplayDimensions() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition
Dim value As System.Object

value = instance.GetDisplayDimensions()
```

C#

```
System.Object GetDisplayDimensions()
```

C++/CLI

```
System.Object^ GetDisplayDimensions();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of [display dimensions](#)

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetDisplayDimensions](#).

### Example

[Get Block Information \(VBA\)](#)

### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetDisplayDimensionCount Method](#)  
[ISketchBlockDefinition::GetDisplayDimensions Method](#)

### Availability

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## GetEllipseCount Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetEllipseCount Method (ISketchBlockDefinition)

Gets the number of ellipses in this block definition.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetEllipseCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Integer  
  
value = instance.GetEllipseCount()
```

C#

```
System.int GetEllipseCount()
```

C++/CLI

```
System.int GetEllipseCount();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Number of ellipses

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetEllipseCount](#).

## Remarks

Call this method before calling [ISketchBlockDefinition::IGetEllipses](#) to get the size of the array for that method.

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetEllipses Method](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## GetEllipses Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetEllipses Method (ISketchBlockDefinition)

Gets the information about all of the ellipses in this block definition.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetEllipses() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Object  
  
value = instance.GetEllipses()
```

C#

```
System.object GetEllipses()
```

C++/CLI

```
System.Object^ GetEllipses();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Array containing an array of doubles (see **Remarks**)

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetEllipses](#).

## Remarks

The return values are in an array of doubles:

[ *Color*, *LineType*, *LineFont*, *LineWidth*, *LayerID*, *LayerOverride*, *StartPt*[3], *EndPt*[3], *CenterPt*[3], *MajorPt*[3], *MinorPt*[3], *Direction* ... ]

where:

|                    |                                                                                                                                    |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <i>Color</i> :     | COLORREF returned as an integer. Return value could be 0 or -1 for default color.                                                  |
| <i>LineType</i> :  | Line type. Valid returns are defined in <a href="#">swLineTypes_e</a> . A lineType is a combination of a lineStyle and lineWeight. |
| <i>LineFont</i> :  | Value is used for polyline font information. This value can be used as an input to the GetLineFontInfo2 and GetLineFontName2.      |
| <i>LineWidth</i> : | Integer value defining the line width. Valid width values as defined in <a href="#">swLineWeights_e</a> .                          |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>LayerID</i> :       | Integer value indicating which layer holds this entity. Obtain the <a href="#">ILayer</a> object by passing this integer value to <a href="#">ILayerMgr::GetLayerById</a> or <a href="#">ILayerMgr::!GetLayerById</a> .                                                                                                                                                                                                                                                                      |
| <i>LayerOverride</i> : | Integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer. |
| <i>StartPt[3]</i> :    | Array of 3 doubles (X,Y,Z) describing the ellipse start point                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <i>EndPt[3]</i> :      | Array of 3 doubles (X,Y,Z) describing the ellipse end point. If the ellipse is closed, then this will be the same point as the StartPt.                                                                                                                                                                                                                                                                                                                                                      |
| <i>CenterPt[3]</i> :   | Array of 3 doubles (X,Y,Z) describing the ellipse center point.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <i>MajorPt[3]</i> :    | Array of 3 doubles (X,Y,Z) describing a point on the ellipse and on the major axis.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <i>MinorPt[3]</i> :    | Array of 3 doubles (X,Y,Z) describing a point on the ellipse and on the minor axis.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <i>Direction</i> :     | -1 for clockwise, +1 for counterclockwise                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

This set of data repeats for each ellipse in the sketch. The size of the array is (NumEllipses \* 22).

#### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetEllipseCount Method](#)  
[ISketchBlockDefinition::!GetEllipses Method](#)

#### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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### GetFeature Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetFeature Method (ISketchBlockDefinition)

Gets the feature for this block definition.

#### .NET Syntax

Visual Basic (Declaration)

```
Function GetFeature() As Feature
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As Feature
```

```
value = instance.GetFeature()
```

C#

```
Feature GetFeature()
```

C++/CLI

```
Feature^ GetFeature();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

**Return Value**

[Feature](#)

#### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetFeature](#).

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## GetInstanceCount Method (ISketchBlockDefinition)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetInstanceCount Method (ISketchBlockDefinition)

Gets the number of block instances of this block definition.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetInstanceCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Integer
```

```
value = instance.GetInstanceCount()
```

C#

```
System.int GetInstanceCount()
```

C++/CLI

```
System.int GetInstanceCount();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Number of block instances

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetInstanceCount](#).

## Example

[Get and Set Blocks in Any Document \(VBA\)](#)  
[Get Blocks in Drawing \(VBA\)](#)

## Remarks

Call this method before calling [ISketchBlockDefinition::IGetInstances](#) to get the size of the array for that method.

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetInstances Method](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

## GetInstances Method (ISketchBlockDefinition)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetInstances Method (ISketchBlockDefinition)

Gets all of the block instances of this block definition.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetInstances() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Object  
  
value = instance.GetInstances()
```

C#

```
System.object GetInstances()
```

C++/CLI

```
System.Object^ GetInstances();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of all of the [block instances](#)

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetInstances](#).

### Example

[Get and Set Blocks Data in Any Document \(VBA\)](#)  
[Get Block Information \(VBA\)](#)  
[Get Blocks in Drawing \(VBA\)](#)

### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetInstanceCount Method](#)  
[ISketchBlockDefinition::GetInstances Method](#)

### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

## GetLineCount Method (ISketchBlockDefinition)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetLineCount Method (ISketchBlockDefinition)

Gets the number of lines in this block definition.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetLineCount() As System.Integer
```

Visual Basic (Usage)

```
Dim instance As ISketchBlockDefinition  
Dim value As System.Integer  
  
value = instance.GetLineCount()
```

C#

```
System.int GetLineCount()
```

C++/CLI

```
System.int GetLineCount();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Number of lines

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetLineCount](#).

### Example

[Get Block Instance in Part or Assembly \(VBA\)](#)

### Remarks

Call this method before calling [ISketchBlockDefinition::IGetLines](#) to get the size of the array for that method.

### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::IGetLines Method](#)

### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## GetLines Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetLines Method (ISketchBlockDefinition)

Gets information about all of the lines in this block definition.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetLines() As System.Object
```

Visual Basic (Usage)

```
Dim instance As ISketchBlockDefinition  
Dim value As System.Object  
  
value = instance.GetLines()
```

C#

```
System.object GetLines()
```

C++/CLI

```
System.Object^ GetLines();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of doubles (see **Remarks**)

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetLines](#).

### Remarks

The return value is the following array of doubles:

[ *Color*, *Type*, *Line Font*, *Line Width*, *Layer ID*, *Layer Override*, [*Start*], [*End*] ]

where all data values are returned as doubles:

*Color* = COLORREF returned as an integer. Return value could be 0 or -1 for default color.

*Type* = line type. Valid returns as defined in [swLineTypes\\_e](#). A lineType is a combination of a lineStyle and lineWeight.

*Line Font* = line style. Valid line styles as defined in [swLineStyles\\_e](#).

*Line Width* = integer value defining the line width. Valid width values as defined in [swLineWeights\\_e](#).

*Layer ID* = integer value indicating which layer holds this entity. Obtain the [ILayer](#) object by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::!GetLayerById](#).

*Layer Override* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride is returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

*Start*[3] = array of 3 doubles (X,Y,Z) describing the line start point.

*End*[3] = array of 3 doubles (X,Y,Z) describing the line end point.

This set of data repeats for each line in the sketch. The number of doubles returned is (lineCount \* 12). To determine the number of lines in the sketch, use [ISketch::GetLineCount2](#).

### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetLineCount Method](#)  
[ISketchBlockDefinition::!GetLines Method](#)

### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## GetNoteCount Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetNoteCount Method (ISketchBlockDefinition)

Gets the number of notes in this block definition.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetNoteCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition
Dim value As System.Integer

value = instance.GetNoteCount()
```

C#

```
System.int GetNoteCount()
```

C++/CLI

```
System.int GetNoteCount();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Number of notes

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetNoteCount](#).

### Remarks

Call this method before calling [ISketchBlockDefinition::IGetNotes](#) to get the size of the array for that method.

### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[SketchBlockDefinition::GetNotes Method](#)

### Availability

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SOLIDWORKS API Help

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## GetNotes Method (ISketchBlockDefinition)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetNotes Method (ISketchBlockDefinition)

Gets the notes in this block definition.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetNotes() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition
Dim value As System.Object

value = instance.GetNotes()
```

C#

```
System.object GetNotes()
```

C++/CLI

```
System.Object^ GetNotes();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of [notes](#)

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetNotes](#).

## Example

[Get Block Information \(VBA\)](#).

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetNoteCount Method](#)  
[ISketchBlockDefinition::IGetNotes Method](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## GetParabolaCount Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetParabolaCount Method (ISketchBlockDefinition)

Gets the number of parabolas in this block definition.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetParabolaCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Integer
```

```
value = instance.GetParabolaCount()
```

C#

```
System.int GetParabolaCount()
```

C++/CLI

```
System.int GetParabolaCount();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Number of parabolas

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetParabolaCount](#).

## Remarks

Call this method before calling [ISketchBlockDefinition::IGetParabolas](#) to get the size of the array for that method.

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## GetParabolas Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetParabolas Method (ISketchBlockDefinition)

Gets information about all of the parabolas in this block definition.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetParabolas() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Object
```

```
value = instance.GetParabolas()
```

C#

```
System.object GetParabolas()
```

C++/CLI

```
System.Object^ GetParabolas();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Array of doubles (see **Remarks**)

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetParabolas](#).

## Remarks

The return values are in an array of doubles:

[ *Color*, *LineType*, *LineStyleIndex*, *LineWidth*, *LayerID*, *LayerOverride*, *StartPt*[3], *EndPt*[3], *FocusPt*[3], *ApexPt*[3] ]

where:

*Color* = COLORREF returned as an integer. Return value could be 0 or -1 for default color.

*LineType* = line type. Valid returns as defined in [swLineTypes.e](#). A *LineType* is a combination of a line style and line weight.

*LineStyleIndex* = line style. Valid line styles as defined in [swLineStyles.e](#).

*LineWidth* = integer value defining the line width. Valid width values as defined in [swLineWeights.e](#).

*LayerID* = integer value indicating which layer holds this entity. Obtain the [ILayer](#) object by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::!GetLayerById](#).

*LayerOverride* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if *LayerOverride* was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

*StartPt*[3] = array of 3 doubles (X,Y,Z) describing the parabola start point.

*EndPt*[3] = array of 3 doubles (X,Y,Z) describing the parabola end point.

*FocusPt*[3] = array of 3 doubles (X,Y,Z) describing the parabola focus point.

*ApexPt*[3] = array of 3 doubles (X,Y,Z) describing the parabola apex point.

This set of data repeats for each parabola in the sketch. The size of the array is (NumParabolas \* 18).

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetParabolaCount Method](#)  
[ISketchBlockDefinition::IGetParabolas Method](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## GetSketch Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetSketch Method (ISketchBlockDefinition)

Gets the underlying sketch of this block definition.

## .NET Syntax

### Visual Basic (Declaration)

```
Function GetSketch() As Sketch
```

### Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As Sketch
```

```
value = instance.GetSketch()
```

### C#

```
Sketch GetSketch()
```

### C++/CLI

```
Sketch^ GetSketch();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Underlying [sketch](#)

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetSketch](#).

## Remarks

The returned sketch defines this block. The sketch does not appear in the FeatureManager tree.

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetSplineCount Method (ISketchBlockDefinition)

[See Also](#)



Gets the number of splines in this block definition.

## Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetSplineCount( _
    ByRef PointCount As System.Integer _
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As ISketchBlockDefinition
Dim PointCount As System.Integer
Dim value As System.Integer

value = instance.GetSplineCount(PointCount)
```

C#

```
System.int GetSplineCount(
    out System.int PointCount
)
```

C++/CLI

```
System.int GetSplineCount(
    & [Out] System.int PointCount
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*PointCount*  
Number of points (see **Remarks**)

### Return Value

Number of splines in the block definition

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetSplineCount](#).

## Remarks

Call this method before calling [ISketchBlockDefinition::ISketchSplines2](#). PointCount is the value use to determine the size of the array for this method.

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[GetSplines2 Method](#)  
[ISketchBlockDefinition::GetSplineInterpolateCount Method](#)  
[ISketchBlockDefinition::GetSplineParams Method](#)  
[ISketchBlockDefinition::GetSplineParamsCount Method](#)  
[ISketchBlockDefinition::GetSplinesInterpolate Method](#)  
[ISketchBlockDefinition::ISketchSplineParams Method](#)  
[ISketchBlockDefinition::ISketchSplinesInterpolate Method](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

## GetSplineInterpolateCount Method (ISketchBlockDefinition)

[See Also](#)



Gets the number of splines in the sketch block definition and the size of array required to hold the data for the interpolation of these splines.

## Visual Basic Syntax

### Visual Basic (Declaration)

```
Function GetSplineInterpolateCount( _  
    ByRef ArraySize As System.Integer _  
) As System.Integer
```

### Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim ArraySize As System.Integer  
Dim value As System.Integer  
  
value = instance.GetSplineInterpolateCount(ArraySize)
```

### C#

```
System.int GetSplineInterpolateCount(  
    out System.int ArraySize  
)
```

### C++/CLI

```
System.int GetSplineInterpolateCount(  
& [Out] System.int ArraySize  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

##### *ArraySize*

Size of the array needed to hold the spline interpolation data information

#### Return Value

Number of splines in this sketch block definition

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetSplineInterpolateCount](#).

## Remarks

The ArraySize value is intended to be used as the input ArraySize argument to [ISketchBlockDefinition::IGetSplinesInterpolate](#).

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetSplineCount Method](#)  
[ISketchBlockDefinition::GetSplineParams Method](#)  
[ISketchBlockDefinition::GetSplineParamsCount Method](#)  
[ISketchBlockDefinition::GetSplines2 Method](#)  
[ISketchBlockDefinition::GetSplinesInterpolate Method](#)  
[ISketchBlockDefinition::IGetSplineParams Method](#)  
[ISketchBlockDefinition::IGetSplines2 Method](#)

## Availability

SOLIDWORKS 2007 SP2, Revision Number 15.2

SOLIDWORKS API Help

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## GetSplineParams Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetSplineParams Method (ISketchBlockDefinition)

Gets all the parameters of the splines in the sketch block definition.

## Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetSplineParams() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Object
```

```
value = instance.GetSplineParams()
```

C#

```
System.Object GetSplineParams()
```

C++/CLI

```
System.Object^ GetSplineParams();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Array of doubles, containing spline parameters (see **Remarks**)

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetSplineParams](#).

## Remarks

The return value is an array of doubles containing data for all the splines in the sketch block definition.

The first two array elements for each spline contain 4 integer values holding information that describes the rest of the data in that splines parameters:

| Spline Element | Packed Data        |                 |
|----------------|--------------------|-----------------|
|                | low part           | high part       |
| 0              | <i>Dim</i>         | <i>Order</i>    |
| 1              | <i>nCtrlPoints</i> | <i>Periodic</i> |

where:

- *Dim* is the number of dimensions the spline is defined in
- *Order* is the order of the spline
- *nCtrlPoints* is the number of control points
- *Periodic* is 1 for a closed spline or 0 for an open spline

The number of knots depends on whether the spline is periodic or not:

Periodic:                    numKnots = nCtrlPoints + 1

Non-Periodic:            numKnots = nCtrlPoints + Order

The last three array elements for each spline contain 5 integer values holding style and layer information:

| Spline Element | Packed Data  |                  |
|----------------|--------------|------------------|
|                | low part     | high part        |
| i              | <i>Color</i> | <i>lineStyle</i> |

|     |                      |                 |
|-----|----------------------|-----------------|
| i+1 | <i>lineWidth</i>     | <i>Layer</i>    |
| i+2 | <i>layerOverride</i> | <i>Not used</i> |

where:

- *i* is the index following the last Knot or  $[2 + \text{numKnots} + \text{numControlPointDoubles} * \text{Dim}]$
- *Color* is the COLORREF value describing the color used for the *ith* spline
- *lineStyle* is the line style used for the *ith* spline. Valid values can be found in the [swLineStyles\\_e](#) enumeration
- *lineWidth* is line width used for the *ith* spline. Valid values can be found in the [swLineWeights\\_e](#) enumeration
- *Layer* is an integer index to the layer that the *ith* spline is on
- *layerOverride* is integer with bit flags set to determine which properties, if any, have been overridden or should be overridden.

Therefore, the size of the data for each spline is given by:

$$2 + \text{numKnots} + \text{numControlPointDoubles} * \text{Dim} + 3$$

The ControlPoint data (in the sketch coordinate system) follows the 2 packed data elements, the Knot points, and, finally, the last 3 packed data elements. Subsequent splines follow one another in the array.

[ *packedDouble1*, *packedDouble2*, *ControlPoint1*[*Dimension elements*], *ControlPoint2*[*Dimension elements*],... *knot1*, *knot2*,..., *packedDouble3*, *packedDouble4*, *packedDouble5*, ]

For information about unpacking double arrays into integer pairs, see:

- [Unpacking Double Arrays into Integer Pairs in Visual Basic.NET and Visual Basic](#)
- [Unpacking Double Arrays into Integer Pairs in C++](#)
- [Unpacking Double Arrays into Integer Pairs in C#](#)

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetSplineParamsCount Method](#)  
[ISketchBlockDefinition::IGetSplineParams Method](#)  
[ISketchBlockDefinition::GetSplineCount Method](#)  
[ISketchBlockDefinition::GetSplineInterpolateCount Method](#)  
[ISketchBlockDefinition::GetSplines2 Method](#)  
[ISketchBlockDefinition::GetSplinesInterpolate Method](#)  
[ISketchBlockDefinition::IGetSplines2 Method](#)  
[ISketchBlockDefinition::IGetSplinesInterpolate Method](#)

## Availability

SOLIDWORKS 2007 SP2, Revision Number 15.2

SOLIDWORKS API Help

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## GetSplineParamsCount Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetSplineParamsCount Method (ISketchBlockDefinition)

Gets the number of splines in the sketch block definition and the size of array required to hold the data for the parameters of these splines.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetSplineParamsCount( _
    ByRef ArraySize As System.Integer _
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As ISketchBlockDefinition
Dim ArraySize As System.Integer
Dim value As System.Integer
```

```
value = instance.GetSplineParamsCount(ArraySize)
```

C#



```
System.int GetSplineParamsCount(  
    out System.int ArraySize  
)
```

C++/CLI

```
System.int GetSplineParamsCount(  
& [Out] System.int ArraySize  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*ArraySize*  
Size of the array needed to hold the spline parameters data

#### Return Value

Number of splines in this sketch block definition

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetSplineParamsCount](#).

### Remarks

The ArraySize value is intended to be used as the input ArraySize argument to [ISketchBlockDefinition::IGetSplineParams](#).

### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::IGetSplineParams Method](#)  
[ISketchBlockDefinition::GetSplineParams Method](#)  
[ISketchBlockDefinition::GetSplineCount Method](#)  
[ISketchBlockDefinition::GetSplineInterpolateCount Method](#)  
[ISketchBlockDefinition::GetSplines2 Method](#)  
[ISketchBlockDefinition::GetSplinesInterpolate Method](#)  
[ISketchBlockDefinition::IGetSplines2 Method](#)  
[ISketchBlockDefinition::IGetSplinesInterpolate Method](#)

### Availability

SOLIDWORKS 2007 SP2, Revision Number 15.2

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## GetSplines Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetSplines Method (ISketchBlockDefinition)

Obsolete. Superseded by [ISketchBlockDefinition::GetSplines2](#) and [ISketchBlockDefinition::IGetSplines2](#).

### .NET Syntax

Visual Basic (Declaration)

```
Function GetSplines() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Object
```

```
value = instance.GetSplines()
```

C#

```
System.object GetSplines()
```

C++/CLI

```
System.Object^ GetSplines();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetSplines](#).

#### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)

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### GetSplines2 Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetSplines2 Method (ISketchBlockDefinition)

Gets the spline points by tessellation instead of by interpolation as is done by [ISketchBlockDefinition::GetSplinesInterpolate](#) and [ISketchBlockDefinition::IGetSplinesInterpolate](#).

#### .NET Syntax

Visual Basic (Declaration)

```
Function GetSplines2() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Object
```

```
value = instance.GetSplines2()
```

C#

```
System.object GetSplines2()
```

C++/CLI

```
System.Object^ GetSplines2();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of doubles (see [Remarks](#))

#### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetSplines2](#).

#### Remarks

Format of return values is an array of doubles with the format:

**[ [ *Color*, *LineType*, *LineStyleIndex*, *LineWidth*, *LayerID*, *LayerOverride*, *NumSplinePoints*, [*x,y,z*] ] ]**

where:

- Color* = COLORREF returned as an integer. Return value could be 0 or -1 for default color.
- LineType* = line type. Valid returns as defined in [swLineTypes e](#). A lineType is a combination of a lineStyle and lineWeight.
- LineStyleIndex* = line style. Valid line styles as defined in [swLineStyles e](#).
- LineWidth* = integer value defining the line width. Valid width values as defined in [swLineWeights e](#).
- LayerID* = integer value indicating which layer holds this entity. Obtain the [ILayer](#) object by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::IGetLayerById](#).
- LayerOverride* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties.

If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

This complete set of data repeats itself for each spline found in the sketch block definition. For each spline, the array returned contains the color, line type, number of spline points in the spline, and X,Y,Z value for each of those points. Therefore, the [x,y,z] parameter is an array of NumSplinePoints, which can vary in size from spline to spline.

The [x,y,z] points for each spline are not the same as the points used to generate the spline. This method tessellates the spline based on the display quality and places points along the spline appropriately.

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetSplineCount Method](#)  
[ISketchBlockDefinition::IGetSplines2 Method](#)  
[ISketchBlockDefinition::GetSplineInterpolateCount Method](#)  
[ISketchBlockDefinition::GetSplineParams Method](#)  
[ISketchBlockDefinition::GetSplineParamsCount Method](#)  
[ISketchBlockDefinition::IGetSplineParams Method](#)

## Availability

SOLIDWORKS 2007 SP2, Revision Number 15.2

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## GetSplinesInterpolate Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetSplinesInterpolate Method (ISketchBlockDefinition)

Gets all of the parameters of the splines by interpolation instead of by tessellation as is done by [ISketch::GetSplines2](#) and [ISketch::IGetSplines2](#).

## .NET Syntax

Visual Basic (Declaration)

```
Function GetSplinesInterpolate() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Object  
  
value = instance.GetSplinesInterpolate()
```

C#

```
System.Object GetSplinesInterpolate()
```

C++/CLI

```
System.Object^ GetSplinesInterpolate();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Array of doubles (see [Remarks](#))

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetSplinesInterpolate](#).

## Remarks

The return value is an array of doubles formatted as:

[ [NumSplinePoints, [x, y, z] ], ]

This complete set of data repeats itself for each spline found in the sketch block definition. For each spline, the array returned contains the number of spline points in the spline, and the X,Y,Z value for each of those points.

The  $[x,y,z]$  parameter is an array of NumSplinePoints. For example, if your sketch block definition has two splines and each spline has three points, then the data would be in the following format:

[ 3, x1\_1, y1\_1, z1\_1, x2\_1, y2\_1, z2\_1, x3\_1, y3\_1, z3\_1, 3, x1\_2, y1\_2, z1\_2, x2\_2, y2\_2, z2\_2, x3\_2, y3\_2, z3\_2 ]

The  $[x,y,z]$  points for each spline are the same points used to generate the spline.

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetSplineInterpolateCount Method](#)  
[ISketchBlockDefinition::IGetSplinesInterpolate Method](#)  
[ISketchBlockDefinition::GetSplineCount Method](#)  
[ISketchBlockDefinition::GetSplineParams Method](#)  
[ISketchBlockDefinition::GetSplineParamsCount Method](#)  
[ISketchBlockDefinition::IGetSplineParams Method](#)

## Availability

SOLIDWORKS 2007 SP2, Revision Number 15.2

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## GetUserPoints Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetUserPoints Method (ISketchBlockDefinition)

Gets information about all of the user points in this block definition.

## .NET Syntax

Visual Basic (Declaration)

```
Function GetUserPoints() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim value As System.Object
```

```
value = instance.GetUserPoints()
```

C#

```
System.object GetUserPoints()
```

C++/CLI

```
System.Object^ GetUserPoints();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

Array of doubles (see [Remarks](#))

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetUserPoints](#).

## Remarks

The return value is an array of doubles with this format:

[ *Color*, *Line Font*, *Line Width*, *Layer ID*, *Layer Override*, *ptLoc[3]* ]

where:

- Color* = COLORREF returned as an integer. Return value could be 0 or -1 for default color.

- *Line Font* = line style. Valid line styles can be found in the [swLineStyles\\_e](#) enumeration.
  - *Line Width* = integer value defining the line width. Valid width values can be found in the [swLineWeights\\_e](#) enumeration.
  - *Layer ID* = integer value indicating which layer holds this entity. The [ILayer](#) object can be obtained by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::!GetLayerById](#).
  - *Layer Override* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.
  - *ptLoc[3]* = an array of doubles (x, y, z) describing the point location
- This set of data repeats for each user point in the sketch. The size of the array is (NumPts \* 8).

#### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetUserPointsCount Method](#)  
[ISketchBlockDefinition::!GetUserPoints Method](#)

#### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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### GetUserPointsCount Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : GetUserPointsCount Method (ISketchBlockDefinition)

Gets the number of user points in this block definition.

#### .NET Syntax

Visual Basic (Declaration)

```
Function GetUserPointsCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition
Dim value As System.Integer

value = instance.GetUserPointsCount()
```

C#

```
System.int GetUserPointsCount()
```

C++/CLI

```
System.int GetUserPointsCount();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Number of user points

#### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::GetUserPointsCount](#).

#### Remarks

Call this method before calling [ISketchBlockDefinition::!GetUserPoints](#) to get the size of the array for that method.

#### See Also

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## IGetArcs Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : IGetArcs Method (ISketchBlockDefinition)

Gets information about all of the arcs in the block definition.

## .NET Syntax

### Visual Basic (Declaration)

```
Function IGetArcs( _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

### Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim ArraySize As System.Integer  
Dim value As System.Double  
  
value = instance.IGetArcs(ArraySize)
```

### C#

```
System.double IGetArcs(  
    System.int ArraySize  
)
```

### C++/CLI

```
System.double IGetArcs(  
    & System.int ArraySize  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*ArraySize*  
Number of arcs

### Return Value

- in-process, unmanaged C++: Pointer to an array containing an array of doubles (see **Remarks**)
  - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

## Remarks

Call [ISketchBlockDefinition::GetArcCount](#) before calling this method to get the value for ArraySize.

Return value is an array of doubles with the format:

[ *Color*, *Type*, *Line Font*, *Line Width*, *Layer ID*, *Layer Override*, [*Start*], [*End*], [*Center*] *Rotation Dir* ]

where

- *Color* - COLORREF returned as an integer. Return value could be 0 or -1 for default color.
- *Type* - Line type. Valid returns as defined in [swLineTypes\\_e](#). A LineType is a combination of a line style and line weight.
- *Line Font* - Line style. Valid line styles as defined in [swLineStyles\\_e](#) enumeration.
- *Line Width* - Integer value defining the line width. Valid width values as defined in [swLineWeights\\_e](#).
- *Layer ID* - Integer value indicating which layer holds this entity. The [ILayer](#) object can be obtained by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::!GetLayerID](#).

- *Layer Override* - Integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, the color and style have been specifically set for this item and may not match the default values associated with this item's layer.
  - *Start[3]* - An array of 3 doubles (X,Y,Z) describing the start point.
  - *End[3]* - An array of 3 doubles (X,Y,Z) describing the end point. If the arc is closed, then End[3] is the same point as Start.
  - *Center[3]* - An array of 3 doubles (X,Y,Z) describing the center point.
  - *Rotation Dir* - Rotational direction (CW = -1, CCW = 1)
- This set of data repeats for each arc in the sketch. The size of the array is (NumArcs \* 16).

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetArcs Method](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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## IGetDisplayDimensions Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : IGetDisplayDimensions Method (ISketchBlockDefinition)

Gets the display dimensions in this block definition.

## .NET Syntax

### Visual Basic (Declaration)

```
Function IGetDisplayDimensions( _
    ByVal Count As System.Integer _
) As DisplayDimension
```

### Visual Basic (Usage)

```
Dim instance As ISketchBlockDefinition
Dim Count As System.Integer
Dim value As DisplayDimension

value = instance.IGetDisplayDimensions(Count)
```

### C#

```
DisplayDimension IGetDisplayDimensions(
    System.int Count
)
```

### C++/CLI

```
DisplayDimension^ IGetDisplayDimensions(
    & System.int Count
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*Count*

Number of display dimensions

### Return Value

- in-process, unmanaged C++: Pointer to an array of [display dimensions](#)
  - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

## Remarks

Call [ISketchBlockDefinition::GetDisplayDimensionCount](#) before calling this method to get the value of Count.

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetDisplayDimensions Method](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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## IGetEllipses Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : IGetEllipses Method (ISketchBlockDefinition)

Gets the information about all of the ellipses in this block definition.

## .NET Syntax

### Visual Basic (Declaration)

```
Function IGetEllipses( _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

### Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim ArraySize As System.Integer  
Dim value As System.Double  
  
value = instance.IGetEllipses(ArraySize)
```

### C#

```
System.double IGetEllipses(  
    System.int ArraySize  
)
```

### C++/CLI

```
System.double IGetEllipses(  
& System.int ArraySize  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*ArraySize*  
Number of ellipses

### Return Value

- in-process, unmanaged C++: Pointer to an array containing an array of doubles (see **Remarks**)
  - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

## Remarks

Call [ISketchBlockDefinition::GetEllipseCount](#) before calling this method to get the value of ArraySize.

The return values are in an array of doubles:

[ *Color*, *LineType*, *LineFont*, *LineWidth*, *LayerID*, *LayerOverride*, *StartPt*[3], *EndPt*[3], *CenterPt*[3], *MajorPt*[3], *MinorPt*[3], *Direction* ... ]

where:

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Color :</b>        | COLORREF returned as an integer. Return value could be 0 or -1 for default color.                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>LineType :</b>     | Line type. Valid returns are defined in <a href="#">swLineTypes_e</a> . A lineType is a combination of a lineStyle and lineWeight.                                                                                                                                                                                                                                                                                                                                                           |
| <b>LineFont :</b>     | Value is used for polyline font information. This value can be used as an input to the GetLineFontInfo2 and GetLineFontName2.                                                                                                                                                                                                                                                                                                                                                                |
| <b>LineWidth :</b>    | Integer value defining the line width. Valid width values as defined in <a href="#">swLineWeights_e</a> .                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>LayerID:</b>       | Integer value indicating which layer holds this entity. Obtain the <a href="#">ILayer</a> object by passing this integer value to <a href="#">ILayerMgr::GetLayerById</a> or <a href="#">ILayerMgr::!GetLayerById</a> .                                                                                                                                                                                                                                                                      |
| <b>LayerOverride:</b> | Integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer. |
| <b>StartPt[3] :</b>   | Array of 3 doubles (X,Y,Z) describing the ellipse start point                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>EndPt[3] :</b>     | Array of 3 doubles (X,Y,Z) describing the ellipse end point. If the ellipse is closed, then this will be the same point as the StartPt.                                                                                                                                                                                                                                                                                                                                                      |
| <b>CenterPt[3] :</b>  | Array of 3 doubles (X,Y,Z) describing the ellipse center point.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>MajorPt[3] :</b>   | Array of 3 doubles (X,Y,Z) describing a point on the ellipse and on the major axis.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>MinorPt[3] :</b>   | Array of 3 doubles (X,Y,Z) describing a point on the ellipse and on the minor axis.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Direction :</b>    | -1 for clockwise, +1 for counterclockwise                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

This set of data repeats for each ellipse in the sketch. The size of the array is (NumEllipses \* 22).

#### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetEllipses Method](#)

#### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## IGetInstances Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : IGetInstances Method (ISketchBlockDefinition)

Gets all of the block instances of this block definition.

#### .NET Syntax

Visual Basic (Declaration)

```
Function IGetInstances( _
    ByVal Count As System.Integer _
) As SketchBlockInstance
```

Visual Basic (Usage)

```
Dim instance As ISketchBlockDefinition
Dim Count As System.Integer
Dim value As SketchBlockInstance
```

```
value = instance.IGetInstances(Count)
```

C#

```
SketchBlockInstance IGetInstances(
    System.int Count
```

)

C++/CLI

```
SketchBlockInstance^ IGetInstances(  
& System.int Count  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Count*

Number of block instances

#### Return Value

- in-process, unmanaged C++: Pointer to an array of all of the [block instances](#)
  - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

#### Remarks

Call [ISketchBlockDefinition::GetInstanceCount](#) before calling this method to get the value of Count.

#### See Also

[ISketchBlockDefinition Interface](#)

[ISketchBlockDefinition Members](#)

[ISketchBlockDefinition::GetInstances Method](#)

#### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## IGetLines Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : IGetLines Method (ISketchBlockDefinition)

Gets information about all of the lines in this block definition.

#### .NET Syntax

Visual Basic (Declaration)

```
Function IGetLines( _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim ArraySize As System.Integer  
Dim value As System.Double  
  
value = instance.IGetLines(ArraySize)
```

C#

```
System.double IGetLines(  
    System.int ArraySize  
)
```

C++/CLI

```
System.double IGetLines(  
& System.int ArraySize  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Parameters

*ArraySize*

Number of lines

## Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
  - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

## Remarks

Call [ISketchBlockDefinition::GetLineCount](#) before calling this method to get the value for ArraySize.

The return value is the following array of doubles:

[ *Color*, *Type*, *Line Font*, *Line Width*, *Layer ID*, *Layer Override*, [*Start*], [*End*] ]

where all data values are returned as doubles:

*Color* = COLORREF returned as an integer. Return value could be 0 or -1 for default color.

*Type* = line type. Valid returns as defined in [swLineTypes\\_e](#). A lineType is a combination of a lineStyle and lineWeight.

*Line Font* = line style. Valid line styles as defined in [swLineStyles\\_e](#).

*Line Width* = integer value defining the line width. Valid width values as defined in [swLineWeights\\_e](#).

*Layer ID* = integer value indicating which layer holds this entity. Obtain the [ILayer](#) object by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::IGetLayerById](#).

*Layer Override* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride is returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

*Start*[3] = array of 3 doubles (X,Y,Z) describing the line start point.

*End*[3] = array of 3 doubles (X,Y,Z) describing the line end point.

This set of data repeats for each line in the sketch. The number of doubles returned is (lineCount \* 12). To determine the number of lines in the sketch, use [ISketch::GetLineCount2](#).

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetLines Method](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

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## IGetNotes Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : IGetNotes Method (ISketchBlockDefinition)

Gets the notes in this block definition.

## .NET Syntax

Visual Basic (Declaration)

```
Function IGetNotes( _  
    ByVal Count As System.Integer _  
) As Note
```

Visual Basic (Usage)

```
Dim instance As ISketchBlockDefinition  
Dim Count As System.Integer  
Dim value As Note
```



```
value = instance.IGetNotes(Count)
```

C#

```
Note IGetNotes(  
    System.int Count  
)
```

C++/CLI

```
Note^ IGetNotes(  
&    System.int Count  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Count*

Number of notes

#### Return Value

- in-process, unmanaged C++: Pointer to an array of [notes](#)
  - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

#### Remarks

Call [ISketchBlockDefinition::GetNoteCount](#) before calling this method to get the value for Count.

#### See Also

[ISketchBlockDefinition Interface](#)

[ISketchBlockDefinition Members](#)

[ISketchBlockDefinition::GetNotes Method](#)

#### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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## IGetParabolas Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : IGetParabolas Method (ISketchBlockDefinition)

Gets information about all of the parabolas in this block definition.

#### .NET Syntax

Visual Basic (Declaration)

```
Function IGetParabolas( _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim ArraySize As System.Integer  
Dim value As System.Double
```

```
value = instance.IGetParabolas(ArraySize)
```

C#

```
System.double IGetParabolas(  
    System.int ArraySize  
)
```

```
System.double IGetParabolas(
& System.int ArraySize
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*ArraySize*  
Number of parabolas

#### Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
  - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

#### Remarks

Call [ISketchBlockDefinition::GetParabolaCount](#) before calling this method to get value for *ArraySize*.

The return values are in an array of doubles:

[ *Color*, *LineType*, *LineStyleIndex*, *LineWidth*, *LayerID*, *LayerOverride*, *StartPt*[3], *EndPt*[3], *FocusPt*[3], *ApexPt*[3] ]

where:

*Color* = COLORREF returned as an integer. Return value could be 0 or -1 for default color.

*LineType* = line type. Valid returns as defined in [swLineTypes\\_e](#). A *LineType* is a combination of a line style and line weight.

*LineStyleIndex* = line style. Valid line styles as defined in [swLineStyles\\_e](#).

*LineWidth* = integer value defining the line width. Valid width values as defined in [swLineWeights\\_e](#).

*LayerID* = integer value indicating which layer holds this entity. Obtain the [ILayer](#) object by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::IGetLayerById](#).

*LayerOverride* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if *LayerOverride* was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

*StartPt*[3] = array of 3 doubles (X,Y,Z) describing the parabola start point.

*EndPt*[3] = array of 3 doubles (X,Y,Z) describing the parabola end point.

*FocusPt*[3] = array of 3 doubles (X,Y,Z) describing the parabola focus point.

*ApexPt*[3] = array of 3 doubles (X,Y,Z) describing the parabola apex point.

This set of data repeats for each parabola in the sketch. The size of the array is (NumParabolas \* 18).

#### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetParabolas Method](#)

#### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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## IGetSplineParams Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : IGetSplineParams Method (ISketchBlockDefinition)

Gets all the parameters of the splines in the sketch block definition.

#### .NET Syntax

Visual Basic (Declaration)

```
Function IGetSplineParams( _
    ByVal ArraySize As System.Integer _
```

) As System.Double

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition
Dim ArraySize As System.Integer
Dim value As System.Double

value = instance.IGetSplineParams(ArraySize)
```

C#

```
System.double IGetSplineParams(
    System.int ArraySize
)
```

C++/CLI

```
System.double IGetSplineParams(
    & System.int ArraySize
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*ArraySize*

Size of the array needed to hold the spline parameters data

#### Return Value

- in-process, unmanaged C++: Pointer to an array of doubles, containing spline parameters (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

#### Remarks

Call [ISketchBlockDefinition::GetSplineParamsCount](#) before calling this method to get the value for ArraySize.

The return value is an array of doubles containing data for all the splines in the sketch block definition.

The first two array elements for each spline contain 4 integer values holding information that describes the rest of the data in that splines parameters:

| Spline Element | Packed Data        |                 |
|----------------|--------------------|-----------------|
|                | low part           | high part       |
| 0              | <i>Dim</i>         | <i>Order</i>    |
| 1              | <i>nCtrlPoints</i> | <i>Periodic</i> |

where:

- *Dim* is the number of dimensions the spline is defined in
- *Order* is the order of the spline
- *nCtrlPoints* is the number of control points
- *Periodic* is 1 for a closed spline or 0 for an open spline

The number of knots depends on whether the spline is periodic or not:

Periodic:                numKnots = nCtrlPoints + 1

Non-Periodic:        numKnots = nCtrlPoints + Order

The last three array elements for each spline contain 5 integer values holding style and layer information:

| Spline Element | Packed Data |           |
|----------------|-------------|-----------|
|                | low part    | high part |

|     |               |           |
|-----|---------------|-----------|
| i   | Color         | lineStyle |
| i+1 | lineWidth     | Layer     |
| i+2 | layerOverride | Not used  |

where:

- i is the index following the last Knot or  $[2 + \text{numKnots} + \text{numControlPointDoubles} * \text{Dim}]$
- Color is the COLORREF value describing the color used for the *ith* spline
- lineStyle is the line style used for the *ith* spline. Valid values can be found in the [swLineStyles\\_e](#) enumeration
- lineWidth is line width used for the *ith* spline. Valid values can be found in the [swLineWeights\\_e](#) enumeration
- Layer is an integer index to the layer that the *ith* spline is on
- layerOverride is integer with bit flags set to determine which properties, if any, have been overridden or should be overridden.

Therefore, the size of the data for each spline is given by:

$$2 + \text{numKnots} + \text{numControlPointDoubles} * \text{Dim} + 3$$

The ControlPoint data (in the sketch coordinate system) follows the 2 packed data elements, the Knot points, and, finally, the last 3 packed data elements. Subsequent splines follow one another in the array.

[ packedDouble1, packedDouble2, ControlPoint1[Dimension elements], ControlPoint2[Dimension elements],... knot1, knot2,..., packedDouble3, packedDouble4, packedDouble5, ]

For information about unpacking double arrays into integer pairs, see:

- [Unpacking Double Arrays into Integer Pairs in Visual Basic.NET and Visual Basic](#)
- [Unpacking Double Arrays into Integer Pairs in C++](#)
- [Unpacking Double Arrays into Integer Pairs in C#](#)

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[GetSplineParams Method](#)  
[ISketchBlockDefinition::GetSplineCount Method](#)  
[ISketchBlockDefinition::GetSplineInterpolateCount Method](#)  
[ISketchBlockDefinition::GetSplines2 Method](#)  
[ISketchBlockDefinition::GetSplinesInterpolate Method](#)  
[ISketchBlockDefinition::IGetSplines2 Method](#)  
[ISketchBlockDefinition::IGetSplinesInterpolate Method](#)

## Availability

SOLIDWORKS 2007 SP2, Revision Number 15.2

SOLIDWORKS API Help

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## IGetSplines Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : IGetSplines Method (ISketchBlockDefinition)

Obsolete. Superseded by [ISketchBlockDefinition::GetSplines2](#) and [ISketchBlockDefinition::IGetSplines2](#).

## .NET Syntax

Visual Basic (Declaration)

```
Function IGetSplines( _
    ByVal ArraySize As System.Integer _
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As ISketchBlockDefinition
Dim ArraySize As System.Integer
Dim value As System.Double

value = instance.IGetSplines(ArraySize)
```

C#



```
System.double IGetSplines(  
    System.int ArraySize  
)
```

C++/CLI

```
System.double IGetSplines(  
&    System.int ArraySize  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*ArraySize*

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::IGetSplines](#).

### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)

SOLIDWORKS API Help

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## IGetSplines2 Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : IGetSplines2 Method (ISketchBlockDefinition)

Gets the spline points by tessellation instead of by interpolation as is done by [ISketchBlockDefinition::GetSplinesInterpolate](#) and [ISketchBlockDefinition::IGetSplinesInterpolate](#).

### .NET Syntax

Visual Basic (Declaration)

```
Function IGetSplines2( _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim ArraySize As System.Integer  
Dim value As System.Double  
  
value = instance.IGetSplines2(ArraySize)
```

C#

```
System.double IGetSplines2(  
    System.int ArraySize  
)
```

C++/CLI

```
System.double IGetSplines2(  
&    System.int ArraySize  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*ArraySize*  
Number of splines

#### Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)

- VBA, VB.NET, C#, and C++/CLI: Not supported  
See [In-process Methods](#) for details about this type of method.

## Remarks

Call [ISketchBlockDefinition::GetSplineCount](#) before calling this method to get the value for ArraySize.

Format of return values is an array of doubles with the format:

**[[ Color, LineType, LineStyleIndex, LineWidth, LayerID, LayerOverride, NumSplinePoints, [x,y,z] ]]**

where:

- *Color* = COLORREF returned as an integer. Return value could be 0 or -1 for default color.
- *LineType* = line type. Valid returns as defined in [swLineTypes e](#). A lineType is a combination of a lineStyle and lineWeight.
- *LineStyleIndex* = line style. Valid line styles as defined in [swLineStyles e](#).
- *LineWidth* = integer value defining the line width. Valid width values as defined in [swLineWeights e](#).
- *LayerID* = integer value indicating which layer holds this entity. Obtain the [ILayer](#) object by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::IGetLayerById](#).
- *LayerOverride* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties.

If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style = 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.

This complete set of data repeats itself for each spline found in the sketch block definition. For each spline, the array returned contains the color, line type, number of spline points in the spline, and X,Y,Z value for each of those points. Therefore, the [x,y,z] parameter is an array of NumSplinePoints, which can vary in size from spline to spline.

The [x,y,z] points for each spline are not the same as the points used to generate the spline. This method tessellates the spline based on the display quality and places points along the spline appropriately.

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetSplines2 Method](#)  
[ISketchBlockDefinition::GetSplineInterpolateCount Method](#)  
[ISketchBlockDefinition::GetSplineParams Method](#)  
[ISketchBlockDefinition::GetSplineParamsCount Method](#)  
[ISketchBlockDefinition::IGetSplineParams Method](#)

## Availability

SOLIDWORKS 2007 SP2, Revision Number 15.2

SOLIDWORKS API Help

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## IGetSplinesInterpolate Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : IGetSplinesInterpolate Method (ISketchBlockDefinition)

Gets all of the parameters of the splines by interpolation instead of by tessellation as is done by [ISketch::GetSplines2](#) and [ISketch::IGetSplines2](#).

## .NET Syntax

Visual Basic (Declaration)

```
Function IGetSplinesInterpolate( _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim ArraySize As System.Integer  
Dim value As System.Double  
  
value = instance.IGetSplinesInterpolate(ArraySize)
```

C#

```
System.double IGetSplinesInterpolate(  
    System.int ArraySize  
)
```

C++/CLI

```
System.double IGetSplinesInterpolate(  
& System.int ArraySize  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*ArraySize*

Number of elements in the return array

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::IGetSplinesInterpolate](#).

### Remarks

Call [ISketchBlockDefinition::GetSplineInterpolateCount](#) before calling this method to get the value for ArraySize.

The return value is an array of doubles formatted as:

[ [NumSplinePoints, [x, y, z] ], ]

This complete set of data repeats itself for each spline found in the sketch block definition. For each spline, the array returned contains the number of spline points in the spline, and the X,Y,Z value for each of those points.

The [x,y,z] parameter is an array of NumSplinePoints. For example, if your sketch block definition has two splines and each spline has three points, then the data would be in the following format:

[ 3, x1\_1, y1\_1, z1\_1, x2\_1, y2\_1, z2\_1, x3\_1, y3\_1, z3\_1, 3, x1\_2, y1\_2, z1\_2, x2\_2, y2\_2, z2\_2, x3\_2, y3\_2, z3\_2 ]

The [x,y,z] points for each spline are the same points used to generate the spline.

### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetSplinesInterpolate Method](#)  
[ISketchBlockDefinition::GetSplineCount Method](#)  
[ISketchBlockDefinition::GetSplineParams Method](#)  
[ISketchBlockDefinition::IGetSplineParams Method](#)

### Availability

SOLIDWORKS 2007 SP2, Revision Number 15.2

SOLIDWORKS API Help

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## IGetUserPoints Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : IGetUserPoints Method (ISketchBlockDefinition)

Gets information about all of the user points in this block definition.

### .NET Syntax

Visual Basic (Declaration)

```
Function IGetUserPoints( _  
    ByVal ArraySize As System.Integer _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition
Dim ArraySize As System.Integer
Dim value As System.Double

value = instance.IGetUserPoints(ArraySize)
```

C#

```
System.double IGetUserPoints(
    System.int ArraySize
)
```

C++/CLI

```
System.double IGetUserPoints(
    & System.int ArraySize
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*ArraySize*  
Number of user points

#### Return Value

- in-process, unmanaged C++: Pointer to an array of doubles (see **Remarks**)
  - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

#### Remarks

Call [ISketchBlockDefinition::GetUserPointsCount](#) before calling this method to get the value for ArraySize.

The return value is an array of doubles with this format:

[ *Color*, *Line Font*, *Line Width*, *Layer ID*, *Layer Override*, *ptLoc[3]* ]

where:

- *Color* = COLORREF returned as an integer. Return value could be 0 or -1 for default color.
- *Line Font* = line style. Valid line styles can be found in the [swLineStyles\\_e](#) enumeration.
- *Line Width* = integer value defining the line width. Valid width values can be found in the [swLineWeights\\_e](#) enumeration.
- *Layer ID* = integer value indicating which layer holds this entity. The [ILayer](#) object can be obtained by passing this integer value to [ILayerMgr::GetLayerById](#) or [ILayerMgr::IGetLayerById](#).
- *Layer Override* = integer with bit flags set to determine which properties, if any, have been overridden with respect to the Layer default properties. If the bit value is set, then the specific property or properties have been overridden. The bit indicators are: color = 0x1, style : 0x2, and width = 0x4. Therefore, if LayerOverride was returned as 3, you know the color and style have been specifically set for this item and may not match the default values associated with this item's layer.
- *ptLoc[3]* = an array of doubles (x, y, z) describing the point location

This set of data repeats for each user point in the sketch. The size of the array is (NumPts \* 8).

#### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::GetUserPoints Method](#)

#### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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## InsertionPoint Property (ISketchBlockDefinition)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : InsertionPoint Property (ISketchBlockDefinition)

Gets or sets the insertion point for the block definition.

## Visual Basic Syntax

Visual Basic (Declaration)

Property InsertionPoint As MathPoint

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition
Dim value As MathPoint

instance.InsertionPoint = value

value = instance.InsertionPoint
```

C#

```
MathPoint InsertionPoint {get; set;}
```

C++/CLI

```
property MathPoint^ InsertionPoint {
    MathPoint^ get();
    void set ( & MathPoint^ value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

[Insertion point](#) for this block definition

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::InsertionPoint](#).

## Example

[Get and Set Blocks in Any Document \(VBA\)](#)

## Remarks

The insertion point of a block definition is the point around which instances of the block rotate or scale. It is the origin of the sketch containing the geometry that is part of the block definition. This implies that the base point of a block definition is always (0,0,0), because the sketch origin is (0,0). Therefore, get version of SketchBlockDefinition::InsertionPoint always returns (0,0,0).

The BlockDefinition::InsertionPoint values are specified relative to the current base point. However, when the insertion point is changed, it must become the sketch origin again, that is, it must become the (0,0,0) point again. So, if you use the this set/put version of this property and then immediately use get version of this property, it returns (0,0,0).

Changing the insertion point does not affect how instances of the block look in a drawing, so when the base point changes, the sketch geometry in the definition is adjusted to allow for that behavior. This means that if you get the block definition sketch and sketch geometry, the sketch geometry coordinates will be different after the insertion point is changed.

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockInstance::Angle Property](#)  
[ISketchBlockInstance::InstancePosition Property](#)  
[ISketchBlockInstance::LockAngle Property](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : LinkToFile Property (ISketchBlockDefinition)

Gets or sets whether the block definition file is linked to a file.

## Visual Basic Syntax

Visual Basic (Declaration)

```
Property LinkToFile As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition
Dim value As System.Boolean

instance.LinkToFile = value

value = instance.LinkToFile
```

C#

```
System.bool LinkToFile {get; set;}
```

C++/CLI

```
property System.bool LinkToFile {
    System.bool get();
    void set ( & System.bool value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True if the block definition is linked to a file, false if not

## Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::LinkToFile](#).

## Example

[Get and Set Blocks in Any Document \(VBA\)](#)  
[Get Block Information \(VBA\)](#)

## Remarks

The file can be either a native SOLIDWORKS block file (**.sldblk**) or a non-native SOLIDWORKS file (**.dxf** or **.dwg**).

This property indicates whether the block definition is linked to an external file, which you can enable or disable on the block definition without destroying the file name. That is, the file name continues to be stored even if the link is disabled.

Use [ISketchBlockDefinition::FileName](#) to get the name of the file to which a block definition is linked.

You can link a block definition to a **.sldblk** file. However, you cannot save a block definition to a **.sldblk** file by only calling [SketchBlockDefinition::FileName](#) (True) and [SketchBlockDefinition::LinkToFile](#) (Filename). You must first either save the block by selecting it and calling [ISketchBlockDefinition::Save](#), or you can assign the block to an already existing **.sldblk** file.

## See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::MakeSketchBlockFromFile Method](#)  
[ISketchBlockDefinition::MakeSketchBlockFromSelected Method](#)  
[ISketchBlockDefinition::MakeSketchBlockFromSketch Method](#)

## Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

## Save Method (ISketchBlockDefinition)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchBlockDefinition Interface](#) : Save Method (ISketchBlockDefinition)

Saves the block definition.

### .NET Syntax

#### Visual Basic (Declaration)

```
Function Save( _  
    ByVal FileName As System.String _  
) As System.Boolean
```

#### Visual Basic (Usage)



```
Dim instance As ISketchBlockDefinition  
Dim FileName As System.String  
Dim value As System.Boolean  
  
value = instance.Save(FileName)
```

#### C#

```
System.bool Save(  
    System.string FileName  
)
```

#### C++/CLI

```
System.bool Save(  
& System.String^ FileName  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*FileName*

Path, filename, and filename extension of **.sldblk** to which to save the block definition

#### Return Value

True if the block definition is saved, false if not

### Visual Basic for Applications (VBA) Syntax

See [SketchBlockDefinition::Save](#).

### See Also

[ISketchBlockDefinition Interface](#)  
[ISketchBlockDefinition Members](#)  
[ISketchBlockDefinition::FileName Property](#)  
[ISketchBlockDefinition::LinkToFile Property](#)  
[ISketchBlockDefinition::MakeSketchBlockFromFile Method](#)  
[ISketchBlockDefinition::MakeSketchBlockFromSelected Method](#)  
[ISketchBlockDefinition::MakeSketchBlockFromSketch Method](#)

### Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0